3 The variation with displacement of the acceleration of an animal's eardrum is shown in Fig. 3.1.

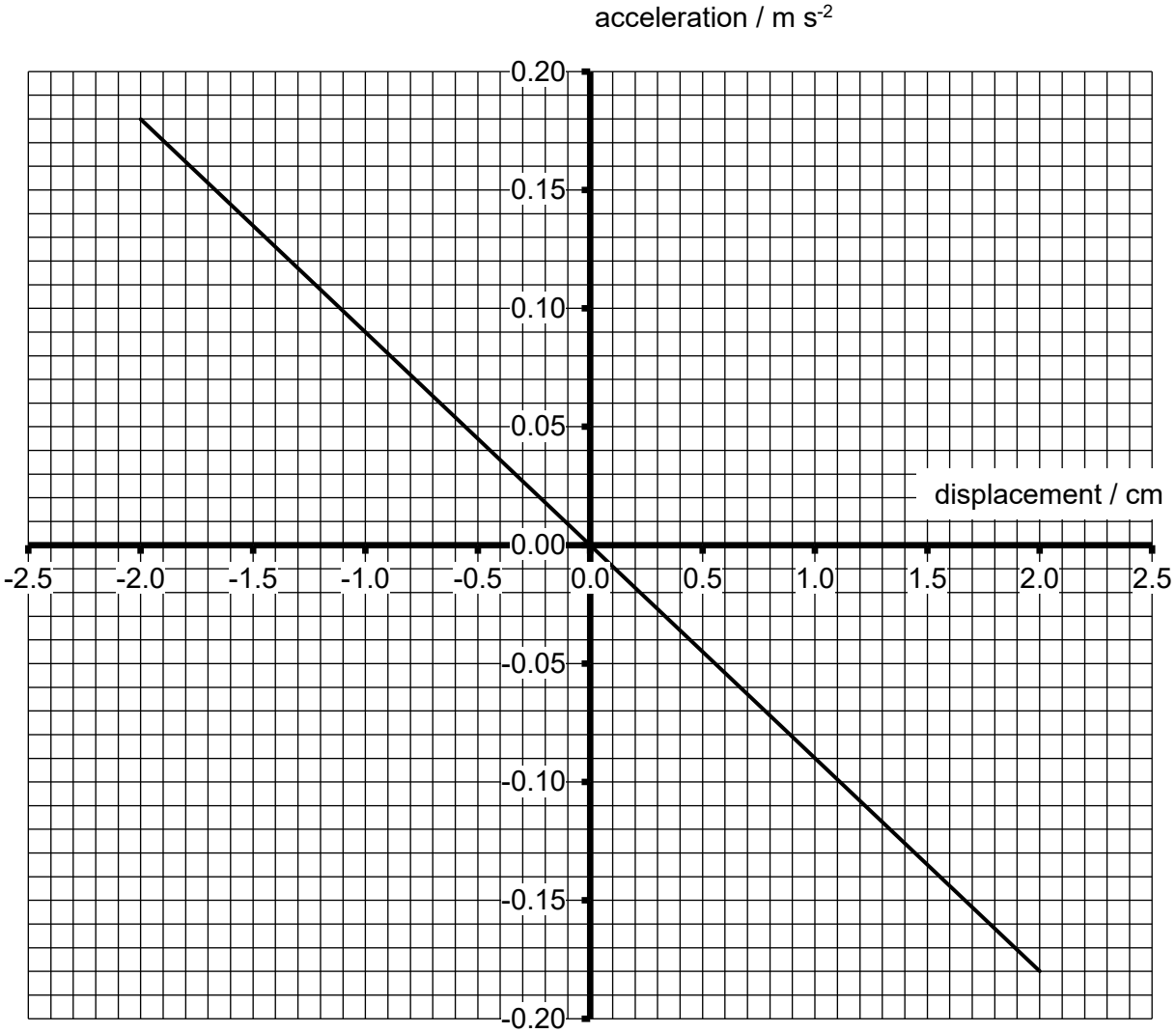

Fig. 3.1

(a) Explain how Fig. 3.1 shows that the motion of the eardrum is simple harmonic.

.....

.....

.....

..... [2]

(b) The period of the oscillation is 2.10 s.

Calculate the time taken for the eardrum to travel a distance of 0.50 cm starting from its maximum displacement towards the equilibrium point.

time taken = …….. s [3]

(c) The mass of the eardrum is 100 g.

Show that the potential energy of the eardrum is 2.5×10^{-5} J when its displacement is 0.75 cm. [2]